

MICRO-MOTION STUDY

➤ *Micro-motion/Therbligs*

- ✓ *Technique of recording and analyzing the timing of basic elements of an operation*
- ✓ *Developing best possible pattern of movement; operator performing @ minimum effort and fatigue, for repeated operations*
- ✓ *Consists of taking motion pictures of the operation with a clock in the picture (or with a video camera running at a known speed)*

MICRO-MOTION STUDY

➤ *Micro-motion/Therbligs*

✓ *The speed of the camera used ranges from 960 to 1000 frames per minute. But faster cameras may be used to study very fast hand motions or complex operations.*

✓ *Micromotion study should be used when it is economical to do so (short cycle highly repetitive operations, large volume production or operation performed by a large number of workers)*

MICRO-MOTION STUDY

➤ *Therbligs*

 Search	 Use
 Find	 Disassemble
 Select	 Inspect
 Grasp	 Preposition
 Hold	 Release Load
 Transport Loaded	 Unavoidable Delay
 Transport Empty	 Avoidable Delay
 Position	 Plan
 Assemble	 Rest

MICRO-MOTION STUDY

➤ *Therbligs*

- *Search (SH) – attempt to find an object using eyes or hand*
- *Find (F) – mental reaction at end of search*
- *Select (ST) – choose among several objects in a group*
- *Grasp (G) – grasp an object*
- *Hold (H) – hold an object*
- *Transport loaded (TL) – move an object with hand and arm*
- *Transport empty (TE) – reach for an object*
- *Position (P) – position object in defined location*
- *Assemble (A) – join two parts*
- *Use (U) – manipulate a tool*
- *Disassemble (DA) – separate multiple parts that were previously joined*
- *Inspect (I) – determine quality of object*
- *Pre-position (PP) – position object for next operation*
- *Release load (RL) – release control of an object*
- *Unavoidable delay (UD) – waiting due to factors beyond worker control*
- *Avoidable delay (AD) – worker waiting*
- *Plan (PN) – decide on an action*
- *Rest (R) – resting to overcome fatigue*

MICRO MOTION STUDY PROCEDURE

1. Filming the operation to be studied
2. Analysing the films
3. Recordings/ presenting the data.

■ **Filming the operation**

- 16mm movie camera
- Other equipments like, reflectors, exposure meter

■ **2)ANALYSING THE FILMS**

- Once the operation over film is viewed through projector.
 1. Film is run at normal speed to get the movement involved
 2. A regular work cycle(process) is selected among the film
 3. Film is run at very slowly for backward purpose.

3) RECORDING / PRESENTING THE DATA

- Once the film analysis over
- Next to prepare SIOM chart for the micro motion of limbs

MICRO-MOTION STUDY

➤ *SIMO chart format:*
(*Simultaneous Motion cycle chart*)

Operation	:		Film No.	:	
Part drawing No.	:		Chart No.	:	
Method	:	 Present/Proposed	Date	:	
Operation No.	:		Charted by:	:	

<i>Wink counter Reading</i>	<i>Left hand description</i>	<i>Therbligs</i>	<i>Time</i>	<i>Time in 200/m</i>	<i>Time</i>	<i>Therbligs</i>	<i>Right hand description</i>

MICRO-MOTION STUDY

- *Provides a permanent record of motion study on films.*
- *A large number of operators can see the procedure at any time even after the completion of motion study work.*
- *Films can easily reveal the difference between the present and the proposed technique.*
- *Films can be demonstrated to large work force at any desired speed.*
- *It provides very accurate time for each operation or motion in comparison to stop watch time study.*

MICRO-MOTION STUDY

- *It helps in making **detailed and accurate analysis** of the prevailing technique.*
- *To study the **activities** of the machine and the operator.*
- *To **impart training** to the workers or operators regarding motion; economy so that unnecessary movement by the workers may be avoided.*
- *To study the **relationship** between the activities of operator and the machine.*
- *To obtain motion time data for **developing synthetic time standards** for various elements.*

MEMO MOTION

(8 mark)

- It is developed by Marvin Mudrel
- It is special type of micro motion,
ACTIVITIES ARE FILMED AT SLOWER
SPEED 60 or 100 frames per second.
- It is best suited for many activities
- For quantity checking purpose in continuous
production activity in company.

PROCEDURE OF MEMO MOTION STUDY

1. Study of operations to be filmed
2. Obtain the film
3. Analysis the film
4. Construct a suitable chart depending up on the nature of the film
5. Improve the methods by using the motion economy

PURPOSE OF MEMO MOTION STUDY AND ADVANTAGES

- To study job involving cycles
- To study team work and create activities
- To study flow of material
- Analyses the material handling
- For collecting motion time data

ADVANTAGES:

- More economical than micro motion study
- Consumes less film
- Film analyses in smaller time than micro motion stud